Summary in Graph

Exam Summary (GO Classes GATE CS| Functions| Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (37) out of 365 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Mathematical Logic

Let f and g be the functions from the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Then the composition of f and g and g and f is given as

A. 6x+7, 6x+11
B. 6x+11, 6x+7
C. 5x+5, 5x+5

D. None of the above

Your Answer: | Correct Answer: A,B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Discrete Mathematics

The following functions all have $\{1, 2, 3, 4, 5\}$ as both their domain and codomain. Which of the following is both injective \& surjective?

A. $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 3 & 3 & 3 & 3 \end{pmatrix}$.

B. $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$.

C. $f(x) = 6 - x$.

D. $f(x) = \begin{cases} x/2 & \text{if } x \text{ is even} \\ (x+1)/2 & \text{if } x \text{ is odd} \end{cases}$.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Consider the function $f : \mathbb{N} \to \mathbb{N}$ given recursively by $f(0) = 1$ and $f(n+1) = 2 \cdot f(n)$. Find $f(10)$.

Your Answer: | Correct Answer: 1024 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

If we define the functions f, g and h that map R into R by :
$f(x) = x^4, g(x) = \sqrt{x^2+1}, h(x) = x^2 + 72$, then the value of the composite functions $ho(gof)$ and $(hog)of$ are given as

A. $x^8 - 71$ and $x^8 - 71$
B. $x^8 - 73$ and $x^8 - 73$
C. $x^8 + 71$ and $x^8 + 71$
D. $x^8 + 73$ and $x^8 + 73$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Suppose $\mathrm{f} : \mathrm{X} \to \mathrm{Y}$ is a function. Which of the following are possible?

A. f is injective but not surjective.
B. f is surjective but not injective.
C. f is surjective and injective.
D. f is neither surjective nor injective.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let X be a finite set. Suppose $\mathrm{f} : \mathrm{X} \rightarrow \mathrm{X}$ is a function. Which of the following are possible?

A. f is injective but not surjective.
B. f is surjective but not injective.
C. f is surjective and injective.
D. f is neither surjective nor injective.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

The functions mapping R into R are defined as :
$f(x) = x^3 - 4x, g(x) = \frac{1}{x^2+1}$ and $h(x) = x^4$.
Then find the value of the following composite functions :
$h_o g(x)$ and $h_o g_o f(x)$

A. $(x^2+1)^4$ and $\left[(x^3-4x)^2+1\right]^4$

B. $(x^2+1)^4$ and $\left[(x^3-4x)^2+1\right]^{-4}$
C. $(x^2+1)^{-4}$ and $\left[(x^2-4x)^2+1\right]^4$
D. $(x^2+1)^{-4}$ and $\left[(x^3-4x)^2+1\right]^{-4}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 04sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

Define $f(n) = \frac{n}{2} + \frac{1-(-1)^n}{4}$ for all $n \in \mathbb{Z}$. Thus, $f : \mathbb{Z} \rightarrow \mathbb{Z}, \mathbb{Z}$ the set of all integers. Which is correct?

A. f is not a function from $\mathbb{Z} \rightarrow \mathbb{Z}$ because $\frac{n}{2} \notin \mathbb{Z}$.
B. f is a function and is not onto but is one-to-one.
C. f is a function and is not onto and not one-to-one
D. f is a function and is onto but not one-to-one.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Let $h = g \circ f : X \rightarrow Z$. Suppose g is one-to-one and onto. Which of the following is FALSE?
A. If f is one-to-one then h is one-to-one and onto.
B. If f is not onto then h is not onto.
C. If f is not one-to-one then h is not one-to-one.
D, If f is one-to-one then h is one-to-one.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let $\mathrm{g} : \mathrm{A} \rightarrow \mathrm{B}$ and $\mathrm{f} : \mathrm{B} \rightarrow \mathrm{C}$ be functions. If $\mathrm{f} \circ \mathrm{g}$ is bijective, then which of the following must be true?

A. g is one to one.
B. g is onto.
C. f is one to one.
D. f is onto.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

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